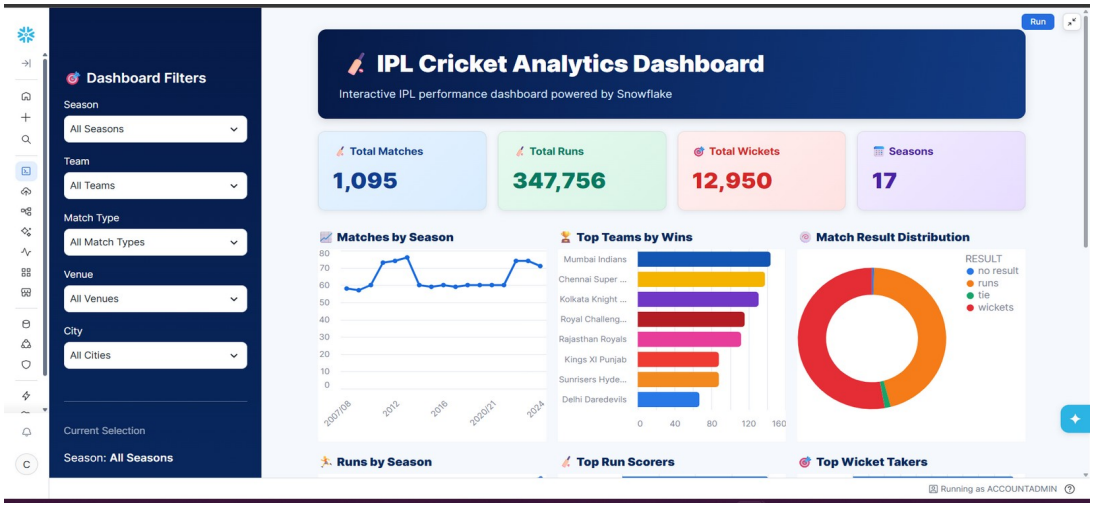
Dashboard Design

Date	29 October 2026
Team ID	Not provided — confirm from MySkillWallet/college
Project Name	EV Market Intelligence Dashboard
Maximum Marks	5 Marks

Activity 1: Interactive and visually appealing dashboards

Creating an effective Streamlit dashboard involves presenting business data stored in Snowflake through clear, interactive, and easy-to-understand visualizations. Students should design the dashboard so that users can quickly understand important KPIs, trends, comparisons, and business insights.

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics





Note: Highlight the major outcomes in form of bullet points

Sample:

Here are five potential outcomes from the dashboard image provided:

1. Mumbai Indians emerge as the strongest team by wins
   Mumbai Indians are shown at the top of the Top Teams by Wins chart, indicating consistently strong performance across IPL seasons.
2. Batting performance is dominated by a few key players
   V. Kohli leads the Top Run Scorers chart, followed by players such as S. Dhawan, RG Sharma, and DA Warner. This highlights the importance of high-performing batsmen.
3. Wicket-taking performance is concentrated among leading bowlers
   YS Chahal ranks highest in the Top Wicket Takers chart, followed by PP Chawla and DJ Bravo, indicating that a relatively small group of bowlers has contributed significantly to wicket-taking.
4. Fielding/toss decisions appear to have an impact on match outcomes
   The Toss Impact and Toss Decision charts show differences between winning after winning the toss and choosing to bat vs. field. This can be used to investigate whether the toss provides a meaningful advantage.
5. IPL match activity and scoring have grown over the seasons
   The Matches by Season and Runs by Season charts show variation and an overall increase in IPL activity and run production across seasons, with particularly high run totals in later seasons.

Overall outcome: The dashboard suggests that IPL success is influenced by a combination of strong teams, elite batting and bowling performers, match/toss decisions, and changing scoring patterns across seasons.

Dashboard Implementation: EV Market Intelligence Dashboard

Built in Tableau Public from ev_market_test.csv (100,000 records). The dashboard combines nine worksheets into a single view. Use a tiled layout and a fixed desktop dashboard size; set each worksheet to Fit → Entire View or Fit Width as appropriate, then check that legends and labels do not overlap.

Design Decisions

• Clear layout: place category sales and top manufacturers in the first row; battery mix and driving-range views next; use the country map and customer ratings for geographic/customer context; finish with model-year trend, capacity-vs-range scatter, and CO_2 savings.

• Appropriate visualizations: sorted horizontal bars for rankings, pie for battery-type record mix, map for country distribution, line chart for model-year trend, and scatter plot for battery capacity vs. range.

• Colour and theming: use a consistent palette; keep battery-type colors consistent across the pie and scatter plot. Avoid using many unrelated colors for ranking charts.

• Filters and interactivity: add only useful filters such as category, battery type, model year, and manufacturing country if the dashboard remains readable. Apply filters to relevant worksheets only.

• Responsive layout: test at the intended presentation size and in Tableau Public's view mode. Avoid floating objects unless there is a specific reason; floating containers commonly cause overlap.

• Labels/tooltips: keep labels on ranking charts where they fit; put secondary details in tooltips. Use clear units (sales units, km, kWh, rating, tons/year).

Major Outcomes

• **Category sales concentration:** Two-Wheeler Scooter has the largest total annual sales units (85,031,983), followed by Sedan (78,308,119) and Compact SUV (76,124,966). Electric Bus is the lowest category total (5,323,646).

• **Leading manufacturers:** SurgeWorks leads the displayed top-10 manufacturers at 4,883,639 annual sales units; CurrentMobility follows at 4,779,221.

• **Battery technology mix:** NMC is the most common battery type (33,836 records), with LFP second (32,101). This chart represents record counts, not market share by sales.

• **Driving range differences:** Luxury Sedan has the highest average range (~549 km), followed by Luxury SUV (~525 km). Two-Wheeler Scooter averages ~105 km.

• **Customer ratings are close:** Category averages are tightly grouped; Crossover is highest at ~4.10 and Electric Truck lowest at ~4.07. Differences are small and should not be overstated.

• **Sales trend by model year:** 2021 has the highest annual sales total (~88.09M); 2025 is lowest (~82.55M) in the provided dataset.

• **Battery capacity and range:** The scatter plot allows users to inspect the relationship and outliers between battery capacity and range, colored by battery type. It is exploratory, not proof of causation.

• **Sustainability metric:** Average reported CO_2 savings are highest for Electric Bus (~9.05 tons/year), Pickup Truck (~9.01), and Electric Truck (~8.96). Interpret these as dataset-reported values, not independently verified real-world impacts.

• **Geographic distribution:** The manufacturing-country map shows the countries represented in the file. Its meaning depends on whether mark size uses record count or summed annual sales; state the measure in the legend/title.

Activity 2: Tableau Public Link

Dashboard URL: [PASTE THE PUBLIC VIEW URL HERE — the published link was not included with the template upload.]

Before submission: open the public link in a signed-out/incognito window, verify that the whole dashboard is visible without scrolling where possible, and capture one full-dashboard screenshot.